

Qns	Answer	Marks
1(a)	$k = [(3)(1.29) (3.3 \times 10^2)^2] / (9.9 \times 10^4)$ $= 4.3$	A1
1(b)	Method 1: $\text{fractional uncertainty} = 2 \times \frac{8}{100} + 0.07 + \frac{0.09}{1.29} = 0.30$ $\Delta k = 0.3 \times 4.2 = 1.3 = 1 \text{ (allow 1 s.f.)}$ Method 1: $k_{\max} = \frac{3(3.3 \times 10^2 \times 1.08)^2 (1.29 + 0.09)}{9.9 \times 10^4 \times 0.93} = 5.712$ $k_{\min} = \frac{3(3.3 \times 10^2 \times 0.92)^2 (1.29 - 0.09)}{9.9 \times 10^4 \times 1.07} = 3.132$ $\Delta k = \frac{1}{2}(5.712 - 3.132) = 1.29 = 1$ $k = 4 \pm 1$	C1 C1 C1 (calc of both kmax and kmin) C1 A1
1(c)	unit of fA = unit of speed unit of A = unit of (speed/f) $\text{unit of } A = \frac{\text{m s}^{-1}}{\text{s}^{-1}}$ = m	M1 A1

